

Deciding Claim Identity Without Trusting Geometry

An Embedding-plus-LLM-Judge architecture for online deliberation, evaluated on adversarial stance triplets

Authors: Tal Yaron, Claude (Anthropic) · **Date:** 2026-07-24

Artefacts: `benchmark/run-sim-e.ts`, `benchmark/compare-sim-e.ts`, `benchmark/results/sim-e*.jsonl`

Companion: `REPORT-2-HYBRID-PREFERENCE-GEOMETRY.md`

Abstract

Deliberation platforms must decide when two free-text statements express the same claim. The standard approach — cosine similarity over sentence embeddings — fails at this task in a specific and dangerous way: it cannot distinguish a paraphrase from a contradiction. On 875 adversarial triplets we measure `text-embedding-3-small` at **0.5%** triplet accuracy, with **99.1%** of opposite-meaning distractors scoring above the production merge threshold of 0.85, and distractors outranking true matches in **863 of 880** cases.

We describe and evaluate **ELJ (Embedding + LLM Judge)**, an architecture in which cosine similarity is demoted from decision rule to candidate ranker and an LLM makes every attach decision. In isolation the judge reaches **95.0%** triplet accuracy at a 2.1% false-merge rate, a 190× improvement over cosine on the same instances. We then evaluate the full architecture under realistic conditions — a *growing corpus* where each statement must find its claim among ~20 live rivals — and report **75.3%** accuracy at an 8.0% false-merge rate, against 55.8% for the best prior configuration.

Three findings generalise beyond this system. (i) **Accuracy lives in the judge prompt, not the architecture:** substituting a plausible hand-written prompt for the production one cost ~40 percentage points with the architecture held constant. (ii) **Hard partitioning destroys recall:** routing statements into topic clusters before searching for their claim caused 59% of all recall loss, because a mis-routed statement is never compared against its twin. (iii) **The remedy is to treat a failed router as a failure, not a discovery** — falling back to an unrestricted search recovers the loss (61.3% → 75.3%, $p = 0.00075$).

1. Problem

1.1 The task

Given a deliberation question and a corpus of participant statements, decide for each new statement whether it expresses a claim already present. The decision drives deduplication, canonical-claim generation, and pro/con structure. Errors are asymmetric and both costly:

- a **missed merge** fragments the claim space and makes synthesis noisy;
- a **false merge** silently files a statement under a claim its author would reject — a representational failure, not merely an accuracy one.

The false-merge case deserves emphasis. If a statement arguing *against* a proposal is filed as expressing it, the platform has misrepresented a participant's position. In a system whose purpose is to surface what people actually think, this is the failure mode that matters most.

1.2 Why embeddings fail

We evaluate on the 875 hard triplets of Blair, Procaccia & Tambe (*Embeddings for Preferences, Not Semantics*). Each triplet is (anchor, **match**, **distractor**): the match shares the anchor's stance in different words; the distractor keeps the anchor's wording and flips the stance.

Table 1 — cosine similarity on hard triplets (`text-embedding-3-small` , production context format, n = 875)

measurement	value
triplet accuracy (raw text)	0.5% [0.2, 1.2]
triplet accuracy (ctx format)	1.8% [1.1, 2.9]
matches ≥ 0.85 (production Pass 1)	70.7%
distractors ≥ 0.85 (false merge at Pass 1)	99.1%
matches ≥ 0.60 (Pass 2)	100%
distractors ≥ 0.60 (false merge at Pass 2)	100%
pipeline-correct (match $\geq 0.6 \wedge$ distractor < 0.6)	0.0% (0/875)

Table 2 — cosine distributions

population	min	median	max
true matches (n = 880)	0.734	0.871	0.962
distractors	0.743	0.969	0.999
English adversarial rewrites (n = 147)	0.738	0.860	0.965
Hebrew paraphrases (n = 125)	0.574	0.702	0.822

The distractor median (0.969) sits *above* the match median (0.871). No threshold separates them. Blair et al. explain why: the cosine margin decomposes into a preference component (stance) and a nuisance component (wording, style, topic), weighted equally. Natural data correlates the two — people who agree tend to phrase things alike — so cosine appears stance-aware until the correlation is deliberately broken.

Two operational consequences follow, and they point in opposite directions:

1. **Cosine is unusable as a decision threshold.** 0.85 admits 99.1% of contradictions.
2. **Cosine is safe and useful as a fetch floor.** At 0.45 no true match was lost in any language tested; the geometry knows what is *relevant* even when it cannot tell *which side*.

ELJ is built on exactly this asymmetry.

1.3 Cross-language failure

Zero of 125 verified Hebrew paraphrases of English anchors reach cosine 0.85; 95% reach 0.60. Any threshold tuned on English silently excludes cross-language duplicates. The same judge scores **125/125** on these pairs. For a multilingual deliberation platform this alone rules out threshold-based merging.

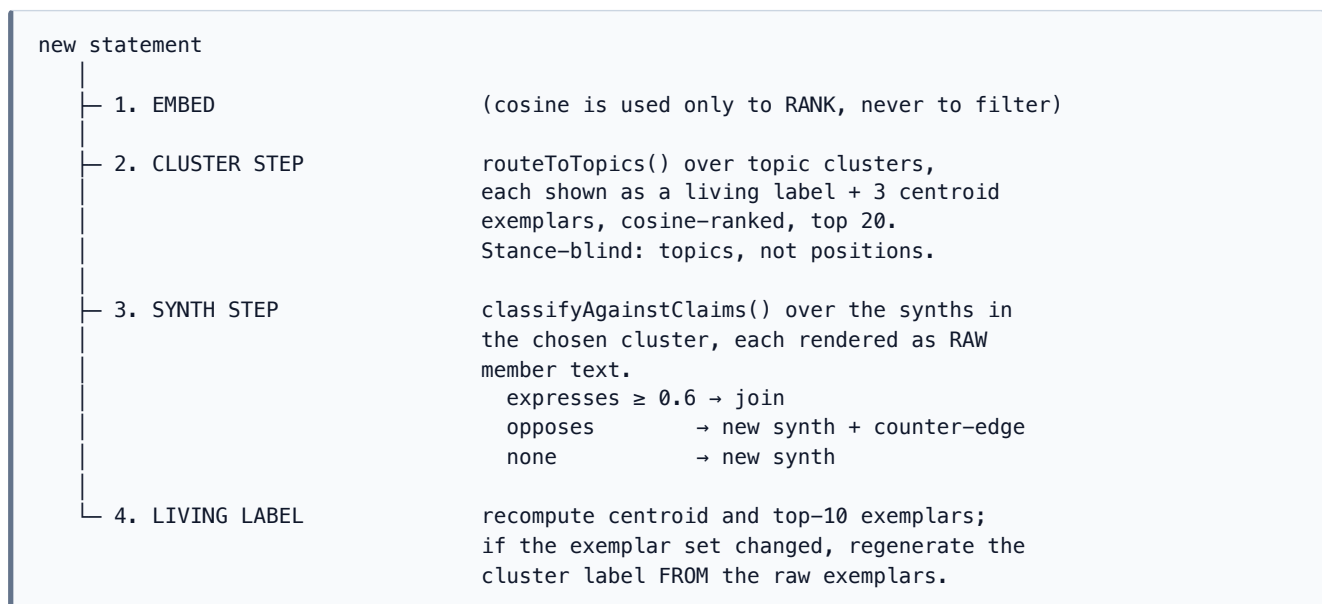
2. Method

2.1 Principle

Cosine proposes; the judge disposes. No attach ever happens on geometry alone, and no candidate is ever excluded by geometry that the judge could have matched.

The judge is a single LLM call (`gpt-4o-mini`) receiving the question, a list of candidate claims rendered as **raw statement text**, and the new statement. It returns `expresses` / `opposes` / `none` with a confidence; attachment requires `expresses` at confidence ≥ 0.6 . An `opposes` verdict is preserved as a counter-edge rather than discarded — "claim X has contradictors" is precisely the pro/con structure synthesis needs.

2.2 Architecture under test (condition E)



Two design decisions carry most of the weight.

Raw text, never summaries. Claims are represented by actual member statements. Condition B2 below shows that compressing a claim to a 5–15-word canonical form costs 25 points of accuracy; the compression discards exactly the stance nuance the task depends on.

Labels are regenerated, not edited. Each label is rewritten *from* the ten centroid-nearest statements rather than revised from the previous label. This prevents drift compounding: a label can never wander further from its members than one regeneration allows.

2.3 Why "same-meaning" and "same-topic" are different questions

The cluster step asks about *subject*; the synth step asks about *claim identity*. Statements on opposite sides of one issue belong in the same cluster and in different synths. This is what produces pro/con structure as a by-product rather than a separate analysis.

3. Experimental setup

Data. The 875 hard triplets of Blair et al., spanning 11 datasets from three platforms (Polis, Remesh, generative social choice). Conditions A–D use all 875; condition E uses a fixed stratified 150-triplet pilot (`results/pilot-ids.json`), proportional per dataset, seeded for reproducibility.

Harness. All judge calls invoke the **production** functions (`classifyAgainstClaims`, `routeToTopics`, `orderClaimsForClassification`) from `functions/src/services/claim-registry-service.ts`. Nothing is reimplemented. §5.1 explains why this constraint is load-bearing.

Condition E protocol — corpus replay. Unlike conditions A–D, which classify isolated triplets, condition E replays a growing corpus per dataset:

- **Phase A (seed):** all 17 anchors stream through the engine in file order, building clusters and synths from nothing.
- **Phase B (probe):** all 34 matches and distractors stream through the same engine in a seeded shuffle, filing into the grown structure.

This is a strictly harder and more realistic task than triplet classification: a match must locate its anchor's synth among ~20 live rivals, and every earlier filing decision changes the structure later statements meet.

Metrics. Per triplet, resolved against the final structure:

metric	definition
<code>synthRecall</code>	match filed into its anchor's synth
<code>clusterRecall</code>	match filed into its anchor's cluster
<code>falseMerge</code>	distractor filed into its anchor's synth
<code>distractorSameCluster</code>	distractor in anchor's cluster (pro/con co-location)
<code>counterEdgeToAnchor</code>	explicit oppose edge recorded
<code>tripletCorrect</code>	$\text{synthRecall} \wedge \neg \text{falseMerge}$

Statistics. Wilson score intervals for proportions; exact McNemar tests for paired configuration comparisons (all configurations see identical triplets).

Noise floor. One configuration (§4.3) differed from baseline by a single structural event yet moved 17 triplets, establishing run-to-run variance at roughly **±5 percentage points** — LLM nondeterminism at temperature 0 plus ordering effects. Differences below this are not interpreted.

4. Results

4.1 The judge in isolation

Table 3 — claim classification, n = 875

condition	claim representation	triplet accuracy	match recall	false merge
A — cosine only	—	0.5%	—	100%
B1 — judge	raw anchor text	95.0% [93.3, 96.2]	96.8%	2.1%
B2 — judge	generated 5–15-word claim	70.1%	79.7%	12.8%
B2E — judge	claim + explanation + exemplar	87.4%	93.3%	6.6%
B2E2 — judge	B2E + stance-caution prompt	88.6%	91.7%	3.8%
D — judge (gpt-4o)	raw anchor text	96.5%	98.3%	1.8%

The judge also names 95.9% of distractors as `opposes` — it does not merely decline to merge, it correctly identifies the relationship, which is what makes counter-edges available for free.

The summary-loss effect (B1 → B2, −24.9pp) is the single largest lever in the table. Representing a claim by its canonical summary rather than raw member text more than doubles the false-merge rate. Enrichment recovers most of it (B2E, B2E2), but never fully.

Embedding gates are safe as floors. Condition A2 applies a cosine gate before the judge: at both 0.45 and 0.60, matches lost at the gate = **0.0%**, and accuracy is identical to ungated B1 (95.0%). The geometry can be trusted to *retrieve*; it cannot be trusted to *decide*.

4.2 The judge under competition

Isolation flatters. With a realistic ~94-claim codebook the same judge must choose among many plausible claims:

Table 4 — full-codebook conditions, n = 875 (CH: n = 613)

condition	triplet accuracy	match → own claim	match → <i>any</i> claim	false merge
C — 94-claim codebook, raw	36.0%	41.3%	74.4%	10.3%
CE — enriched claims	49.6%	62.3%	92.9%	19.3%
CE2 — enriched + stance caution	53.8%	59.0%	89.8%	10.3%
CH — two-hop hierarchical + flat fallback	55.8%	60.8%	92.2%	8.2%

Note the "*match → any claim*" column at 89–93%. Most apparent misses are the match attaching to a different but equivalent claim — defensible behaviour, not error. Competition, not judge quality, accounts for the drop from 95% to ~56%.

CH (55.8%) is therefore the correct baseline for condition E, not B1's 95%.

4.3 The full engine: corpus replay

Table 5 — condition E configurations, n = 150 (identical triplets throughout)

configuration	tripletCorrect	synthRecall	clusterRecall	falseMerge	distSameCluster	counterEdge	clusters
baseline — create cluster on empty routing	61.3% [53, 69]	67.3%	80.7%	8.0%	71.3%	47.3%	56
merge pass (post-seed repair)	66.0% [58, 73]	69.3%	80.0%	4.7%	72.0%	49.3%	58
flat fallback	75.3% [68, 82]	82.7%	98.7%	8.0%	93.3%	68.0%	22
flat fallback + creation guard	79.3% [72, 85]	86.7%	100%	8.0%	100%	71.3%	11
creation guard (cos ≥ 0.70)	82.0% [75, 87]	88.0%	100%	7.3%	100%	72.0%	11

Table 6 — exact McNemar tests

comparison	fixed	broken	p	
baseline → flat fallback	+29	−8	0.00075	***
baseline → creation guard	+34	−3	1.2×10^{-7}	***
baseline → merge pass	+12	−5	0.14	ns
flat fallback → flat + guard	+9	−3	0.15	ns
flat + guard → guard alone	+9	−5	0.42	ns

The flat fallback captures the entire statistically significant gain. Everything beyond it is indistinguishable from it.

Cost. 2.85 LLM calls per statement; mean candidate lists of 9.5 synths and 2.7 clusters — both well inside the short-list regime where the judge's 95% was measured. Approximately \$0.2–0.5 per 1,000 statements at `gpt-4o-mini` rates.

5. Analysis

5.1 Accuracy lives in the judge prompt

The first implementation of the harness used hand-written cluster and synth prompts rather than the production classifier. Measured on the same 14 triplets, with the architecture held constant:

	hand-written prompt	production prompt
tripletCorrect	36%	79%
falseMerge	43%	7%
counterEdge	36%	79%

The production prompt carries two defences the hand-written one lacked:

CAUTION: a statement may reuse an example's exact wording while taking the OPPOSITE stance... Shared phrasing is never evidence of a match. When in doubt between "expresses" and "none", answer "none".

plus the ≥ 0.6 confidence gate. A **plausible-looking substitute prompt cost ~40 percentage points**. This is the largest single effect measured anywhere in this work — larger than every architectural change combined — and it is invisible without an adversarial test set. We retain the failed run (`results/sim-e-handrolled-prompt.jsonl`) as evidence.

Methodological consequence: an evaluation harness must import the production classifier, never approximate it.

5.2 Hard partitioning is the architecture's dominant failure mode

In the baseline, of 49 matches that failed to reach their anchor's synth, **29 (59%) never reached the anchor's cluster at all**. The synth judge — the accurate component — never saw them. Only 20 were genuine synth-step misses.

Fragmentation predicts this loss across datasets:

Pearson r vs clusters-per-statement	
distractorSameCluster	-0.82
clusterRecall	-0.70
synthRecall	-0.53

	clusterRecall	synthRecall	distSameCluster
low fragmentation (≤ 0.10 clusters/statement)	94%	78%	85%
high fragmentation (> 0.15)	66%	58%	51%

One dataset kept all 51 statements in a single cluster and scored 100% on both co-location metrics; another split 51 statements across 11 clusters and fell to 41% / 35%.

Fragmentation also silently destroys the design's structural payoff: `counterEdgeToAnchor` was only 47.3% in the baseline because a counter-edge can only form when the distractor lands in its anchor's cluster. **The pro/con structure is the first thing fragmentation takes.**

5.3 The mechanism: a failed router is not a discovery

The baseline created a new cluster whenever routing returned empty. This is the error. Empirically the statements triggering it were predominantly **matches** — paraphrases of anchors already in the corpus. Seeding 17 anchors produced 1–4 clusters; the probe phase inflated this to 3–12.

The correct rule already existed in the production code (`classifyHierarchical`): empty routing means the *router* failed, not that a new topic was found, so the system falls back to an unrestricted flat search before concluding anything is new. The source comment states the principle exactly:

a gate that filters can misfile; a gate with an ungated second look cannot.

Reproducing this rule in the simulation yields 61.3% → 75.3% ($p = 0.00075$), with 31 statements rescued from spawning spurious clusters. Production further skips routing entirely below 30 claims (`HIERARCHY_MIN_CLAIMS`); our corpora (17–33 synths) sit at that boundary.

5.4 Why the highest-scoring configuration is not recommended

Accuracy on this pilot is **monotone in cluster collapse**:

configuration	clusters	accuracy
baseline	56	61.3%
flat fallback	22	75.3%
flat + guard	11	79.3%
creation guard	11	82.0%

The two best configurations place **every dataset in exactly one cluster**. They do not cluster better; they stop clustering. Since each pilot dataset is a single deliberation question on essentially one topic, the benchmark *cannot penalise this*, whereas a genuinely multi-topic deliberation would. Reported `clusterRecall = 100%` and `distractorSameCluster = 100%` are trivially true at $k = 1$; the flat fallback's 98.7% and 93.3% at $k = 22$ are earned.

We therefore recommend the **flat fallback** — significant, structure-preserving, and already production's rule — and reject the cosine creation guard on this evidence, despite its higher raw score. Adopting it would be fitting to a property of the test set.

5.5 A failed experiment, reported

The merge pass (periodically judging cluster pairs for redundancy and merging) showed no effect ($p = 0.14$), and the experiment was **mis-designed**: it ran once after seeding, when only 1–4 clusters existed, whereas fragmentation occurs during the probe phase that follows. Several datasets generated zero candidate pairs. It also has a structural ceiling — merging clusters deliberately does not merge synths, so a repair pass can never improve `synthRecall` or `tripletCorrect`. **Periodic repair remains untested; this result should not be read as evidence against it.**

6. Limitations

1. **The adversary was built for the opponent we replaced.** The distractors were generated by GPT-4o to defeat *embeddings*: maximum lexical overlap, flipped stance. Nothing in their construction targets an LLM reading the text. Our 95% is measured against an attack our method is structurally well-suited to resist. Distractors generated adversarially *against a judge* — buried negations, scope shifts, conditional hedges, sarcasm — could score materially worse. **95% should not be treated as the judge's ceiling.**

2. **Condition E is $n = 150$** , against $n = 875$ for conditions A–D. Wilson intervals are correspondingly wide ($\pm 7pp$) and the noise floor is $\pm 5pp$.
 3. **Single-topic corpora.** Every pilot dataset is one deliberation question. The benchmark cannot distinguish good coarse clustering from no clustering, which is precisely why §5.4 declines to recommend the highest scorer.
 4. **Scale untested.** Corpora reach 51 statements. The candidate-list growth that motivates hierarchical routing has not been exercised; conclusions about the cluster layer's value at 10^3 – 10^4 statements do not follow from this data.
 5. **Order dependence unmeasured.** Structure depends on arrival order. We used one seeded shuffle; order-shuffled replicates were not run.
 6. **English-dominant evaluation.** Cross-language results (125/125) come from a separate, smaller Hebrew set and were not replayed through the full engine.
 7. **Question text is approximate.** The triplet files carry no question; ours are reconstructed from the source paper's dataset descriptions.
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7. Conclusions

1. **Cosine similarity cannot decide claim identity.** 0.5% triplet accuracy; 99.1% of contradictions above the production merge threshold; 0% pipeline accuracy. This is not a tuning problem — no threshold exists.
2. **An LLM judge can.** 95.0% in isolation, 2.1% false merges, 95.9% of contradictions correctly identified as opposition, and 100% on cross-language paraphrases where cosine reaches 0% — at roughly \$0.2–0.5 per 1,000 statements.
3. **Geometry remains valuable as retrieval.** A 0.45 fetch floor lost no true match in any language. The correct division of labour is *cosine retrieves, the judge decides*.
4. **Represent claims by raw text.** Summarisation costs 25 points and doubles false merges.
5. **Never let a routing layer filter without an ungated second look.** Hard partitioning caused 59% of recall loss; treating empty routing as failure rather than discovery recovers it ($p = 0.00075$).
6. **The judge prompt is the single largest lever.** A plausible substitute cost ~ 40 points with architecture held constant.

The recommended configuration reaches **75.3%** on a corpus-replay task, versus 55.8% for the best prior configuration at a comparable false-merge rate — while preserving the cluster structure that carries pro/con relationships. The dominant remaining loss is candidate ranking: distractors outrank matches 98.1% of the time, so a cosine-ordered candidate list is adversarially ordered. Report 2 addresses that directly.